

UndoProof

Prove the way back

PostgreSQL rollback rehearsal

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The order that disappears

The release succeeds.

A customer places an order.

The rollback erases it.

SQL success does not establish data preservation.

A release contract, executed

Input	Execution	Evidence
Seed + migration SQL	Fresh PGlite databases	Exact statements and errors
Old + new query contracts	Compatibility checks	A result for each release state
New writes + invariant	Rollback, compare, reapply	Before-and-after rows

New writes happen before rollback.

The rollback loses order 104

4 rows before

3 rows after

The rollback SQL succeeds.
The invariant fails.

INSPECT THE RECEIPT

×

Data preservation

⊗ Failed

1 ms

Business data changed: 4 rows before rollback, 3 after. Compare values as well as counts.

Executed SQL

```
SELECT id, customer, total_cents FROM orders ORDER BY id;
```

Before rollback

4 rows

id	customer	total_cents
101	Northstar Studio	24900
102	Juniper Labs	8900
103	Atlas Works	15900
104	New customer order	32900

After rollback

3 rows

id	customer	total_cents
101	Northstar Studio	24900
102	Juniper Labs	8900
103	Atlas Works	15900

Rows missing or changed afterward: 1. Highlighted where visible.

This evidence applies to the contract captured in this run.

One reproducible rehearsal per plan

Plan	Passed checks	Rows after rollback	Engine duration
Unsafe snapshot	9 of 10	3 of 4	2,810.96 ms
Supplied repair	10 of 10	4 of 4	2,808.79 ms

Same invariant. The repair preserves the new order.

Single synthetic CLI run per plan. Engine duration excludes setup and review.
This measures execution, not a percentage productivity improvement.

IBM Bob repository validation

1 Source-derived regression tests

41 checks cover the checkout contract, silent data loss, and the unchanged repair invariant.

2 Real PostgreSQL validation

Bob ran the new suite in the repository using PGLite.

Scoped Bob testing contribution. Sample repairs remain authored examples.

A focused commercial starting point

The first buyer

Platform leads at SaaS teams shipping database changes

Free

MIT local runner + CI

The adoption trigger

The next migration pull request

\$99 / team / month

Proposed shared policies and retained evidence

Pricing hypothesis

A complement to migration tools

Atlas, Liquibase and Flyway already manage migrations and support recovery workflows.

UndoProof focuses the review on old/new query compatibility and data written after deployment.

Tested SQL and fixtures only. Full application behavior, concurrency and production locks remain outside this prototype.

The way back belongs in the release review.

UndoProof

undoproof.web.app

github.com/shi1720/ibm-bob

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